

HW3 — Question 3c

ORF 309

Problem Statement

A bag contains coins: $\frac{2}{3}$ are fair ($P(H) = 0.5$) and $\frac{1}{3}$ are Canadian ($P(H) = 0.65$). One coin is drawn uniformly at random and flipped twice independently. Given that exactly 1 tail was observed, what is the probability the coin is Canadian?

Setup and Notation

Define the following events:

$$\begin{aligned} F &= \{\text{coin drawn is fair}\}, & P(F) &= \frac{2}{3} \\ C &= \{\text{coin drawn is Canadian}\}, & P(C) &= \frac{1}{3} \\ E &= \{\text{exactly 1 tail in 2 flips}\} \end{aligned}$$

We want to compute $P(C \mid E)$.

Step 1: Compute $P(E \mid F)$ and $P(E \mid C)$

Given the coin type, the two flips are independent and identically distributed. So the number of tails $T \mid \text{coin type} \sim \text{Binomial}(2, p_T)$, where p_T is the probability of tails on a single flip.

Fair coin: $p_T = 0.5$

The two sequences giving exactly 1 tail are HT and TH:

$$\begin{aligned} P(E \mid F) &= P(\text{HT} \mid F) + P(\text{TH} \mid F) \\ &= (0.5)(0.5) + (0.5)(0.5) \\ &= 0.25 + 0.25 \\ &= \boxed{0.5} \end{aligned}$$

Equivalently, using the Binomial formula with $n = 2$, $k = 1$, $p_T = 0.5$:

$$P(E | F) = \binom{2}{1} (0.5)^1 (0.5)^1 = 2 \cdot 0.25 = 0.5$$

Canadian coin: $p_T = 0.35$

$$\begin{aligned} P(E | C) &= P(HT | C) + P(TH | C) \\ &= (0.65)(0.35) + (0.35)(0.65) \\ &= 0.2275 + 0.2275 \\ &= \boxed{0.455} \end{aligned}$$

Equivalently, using the Binomial formula with $n = 2$, $k = 1$, $p_T = 0.35$:

$$P(E | C) = \binom{2}{1} (0.35)^1 (0.65)^1 = 2 \cdot (0.35)(0.65) = 2 \cdot 0.2275 = 0.455$$

Step 2: Compute $P(E)$ via the Law of Total Probability

Since F and C are mutually exclusive and exhaustive (every coin is either fair or Canadian):

$$P(E) = P(E | F) P(F) + P(E | C) P(C)$$

Substituting:

$$\begin{aligned} P(E) &= (0.5) \left(\frac{2}{3} \right) + (0.455) \left(\frac{1}{3} \right) \\ &= \frac{1}{3} + \frac{0.455}{3} \\ &= \frac{1 + 0.455}{3} \\ &= \frac{1.455}{3} \\ &\approx 0.485 \end{aligned}$$

Step 3: Apply Bayes' Formula

$$P(C | E) = \frac{P(E | C) P(C)}{P(E)}$$

Substituting:

$$\begin{aligned}
 P(C \mid E) &= \frac{(0.455) \left(\frac{1}{3}\right)}{\frac{1.455}{3}} \\
 &= \frac{\frac{0.455}{3}}{\frac{1.455}{3}} \\
 &= \frac{0.455}{1.455} \\
 &= \frac{91}{291} \\
 &\approx \boxed{0.313}
 \end{aligned}$$

Exact fraction derivation:

$$0.455 = \frac{91}{200}, \quad 1.455 = \frac{291}{200} \quad \implies \quad \frac{0.455}{1.455} = \frac{91/200}{291/200} = \frac{91}{291}$$

Summary

	Fair (F)	Canadian (C)
Prior $P(\cdot)$	$2/3 \approx 0.667$	$1/3 \approx 0.333$
$P(E \mid \cdot)$	0.5	0.455
$P(E \mid \cdot) \cdot P(\cdot)$	0.333	0.152
Posterior $P(\cdot \mid E)$	0.687	0.313

Interpretation. The prior probability of a Canadian coin was 33.3%. After observing exactly 1 tail, the posterior drops slightly to 31.3%. This makes sense: exactly 1 tail in 2 flips is slightly *more* consistent with a fair coin ($P(E \mid F) = 0.5$) than with a Canadian coin ($P(E \mid C) = 0.455$), so the evidence nudges us weakly toward fair — but not by much, since the likelihoods are close.